SMIC Navigation Log

First Proof 2026 Challenge

Claude Sonnet 4.6 — Reasoning Attempt under SMIC Framework

Claude Sonnet 4.6 (Anthropic)

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What this document is. This is a reasoning *log*, not a paper. It records how Claude Sonnet 4.6 attempts the 10 First Proof 2026 challenge problems using the Structural Motion & Interaction Calculus (SMIC) as a navigation framework. No claims about correctness of partial results are made.

SMIC operates via five phases: **Emerge** (construct new structure), **Propagate** (carry an argument forward), **Transform** (switch domain or representation), **Decay** (extremal/descent), **Integrate** (fixed point / saturation). Each step is logged with: operator applied, scale level (0–4), failure channels triggered, and honest verdict.

Where the argument reaches a genuine gap, the gap is logged explicitly. Where the problem statement is underspecified, `C5:Misfit` fires.

#	Short title	Domain	Phase	Scale	Verdict
1	Φ_3^4 shifts on $\mathbb{T}^3$	ANA	**Transform**	3	**[PARTIAL — gap logged]**
2	Twisted RS nonvanishing	ALG	**Transform**	3	**[PARTIAL — gap logged]**
3	Markov chain / interp polys	PROB	**Propagate**	2	**[PARTIAL — gap logged]**
4	Cyclotomic HM inequality	NT	**Integrate**	2	**[OPEN — C5:Misfit]**
5	$\mathcal{O}$-adapted slice filtration	TOP	**Transform**	3	**[PARTIAL — gap logged]**
6	ε-light vertex subsets	COMB	**Decay**	3	**[SKETCH COMPLETE]**
7	Uniform lattices with 2-torsion	TOP	**Emerge**	3	**[PARTIAL — gap logged]**
8	Polyhedral Lagrangian smoothing	GEO	**Transform**	3	**[PARTIAL — gap logged]**
9	Quadri-linear determinant tensors	ALG	**Integrate**	3	**[PARTIAL — gap logged]**
10	Kernelized CP-ALS + PCG	ALG	**Propagate**	3	**[PARTIAL — gap logged]**

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1 P1: Smooth shifts of the Φ^4_3 measure on $\mathbb{T}^3$

Statement. Let $\mathbb{T}^3$ be the 3D unit torus and μ the Φ^4_3 measure on $\mathcal{D}'(\mathbb{T}^3)$. Let $\psi : \mathbb{T}^3 \to \mathbb{R}$ be smooth and define $T_\psi(\phi) = \phi + \psi$. Find necessary and sufficient conditions on ψ such that $(T_\psi)_\# \mu \ll \mu$.

> **SMIC log.** Genesis: domain `ANA`, phase **Transform**.
> Scale ladder entry: Scale 0 (symbolic rewrite). Primary channel monitored: `C6:InvAbsence` (non-Gaussian correction).

1.1 Step 1 — *domain_switch*: Girsanov–Cameron–Martin frame

The Φ^4_3 measure is formally

$$d\mu \;\propto\; \exp\Big(-\int_{\mathbb{T}^3} :\phi^4:\, dx\Big)\, d\mu_{\mathrm{GFF}},$$

where μ_{GFF} is the Gaussian free field measure on $\mathbb{T}^3$. For a pure Gaussian base measure μ_0, the Cameron–Martin theorem gives: $(T_\psi)_\# \mu_0 \ll \mu_0$ iff $\psi \in \mathcal{H}(\mu_0) = H^1(\mathbb{T}^3)$.

SMIC TRANS: switch from QFT language to Girsanov/Cameron–Martin framework.

1.2 Step 2 — *nt_mod_invariant*: Cameron–Martin space for Φ^4_3

Invariant recorded: *ψ must lie in the Cameron–Martin space $\mathcal{H}(\mu)$ of μ. After renormalization, the expected answer is $\mathcal{H}(\mu) = H^1(\mathbb{T}^3)$.*

Formal Radon–Nikodym derivative:

$$\frac{d(T_\psi)_\# \mu}{d\mu} \;= :\exp\Big(I(\psi) - \tfrac{1}{2}\|\psi\|^2_{H^1}\Big):_\mu,$$

where $I(\psi)$ is a Wick-renormalized stochastic integral.

1.3 Step 3 — Scale 3: Novikov-type condition

Absolute continuity holds iff

$$\mathbb{E}_\mu\Big[\exp\big(\tfrac{1}{2}\|I(\psi)\|^2\big)\Big] < \infty.$$

For Φ^4_3, this is controlled by whether $\psi \in H^1(\mathbb{T}^3)$: the $L^2(\mu)$-norm of the Wick exponential is finite precisely in this Sobolev class. Supporting work: Hairer–Ryser–Weber (2012) established quasi-invariance for H^1 shifts in lower-dimensional Φ^4 measures; the $\mathbb{T}^3$ case is expected to follow by stochastic quantization.

> **[PARTIAL — gap logged]**
> **Proposed answer:** $\psi \in H^1(\mathbb{T}^3)$ is necessary and sufficient.
> **Gap:** (i) Renormalization of $I(\psi)$ in 3D is non-trivial (SPDE level). (ii) Equivalence Novikov $\Leftrightarrow \psi \in H^1$ for the non-Gaussian μ is not proved here.
> `C6:InvAbsence` — no closed-form invariant for the non-Gaussian correction term.

2 P2: Nonvanishing test vector for the twisted local Rankin–Selberg integral

Statement. F non-Archimedean, E/F quadratic. π generic irreducible admissible on $\mathrm{GL}_n(E)$. Prove existence of a test vector giving a nonvanishing twisted local Rankin–Selberg integral.

SMIC log. Genesis: domain `ALG` (automorphic/p-adic), phase **Transform**.
Strategic note flags this as high-complexity TRANS. `OpenAI failure note:` "unstable Propagate phase due to misaligned anchor in local field representation" — SMIC interpretation: wrong normalization of additive character ψ_E breaks the Whittaker functional equation.

2.1 Step 1 — *domain_switch*: Whittaker model

Since π is generic it admits a unique Whittaker model $\mathcal{W}(\pi, \psi_E)$ where $\psi_E = \psi_F \circ \mathrm{Tr}_{E/F}$. The twisted local Rankin–Selberg integral:

$$Z(s, W, \Phi) = \int_{N_n(E)\backslash \mathrm{GL}_n(E)} W(g)\, \Phi(e_n g)\, |\det g|^s \, dg.$$

Nonvanishing for *some* (W, Φ) is equivalent to the zeta integral being a nonzero rational function in q^{-s}.

2.2 Step 2 — *repr_switch*: Case split on ramification

Case A (unramified π): Take $W = W^\circ$ (spherical Whittaker) and $\Phi = \mathbf{1}_{M_n(\mathcal{O}_E)}$. Then $Z(s, W^\circ, \Phi) = L(s, \pi, \mathrm{As})$ (Asai L-function), which is nonzero for $\mathrm{Re}(s) \gg 0$. **Test vector for Case A: W°.**

Case B (ramified π, conductor $\mathfrak{f}$): Jacquet–Piatetski-Shapiro–Shalika newform theory provides $W_\mathfrak{f}$ such that $Z(s, W_\mathfrak{f}, \Phi_\mathfrak{f}) = \varepsilon(s, \pi, \mathrm{As}) \cdot L(\cdots) \neq 0$.

2.3 Step 3 — *proof_extremal*: Nonvanishing via functional equation

Local functional equation:

$$Z(1-s, \widetilde{W}, \hat{\Phi}) = \omega_\pi(-1)^n \, \gamma(s, \pi, \mathrm{As}, \psi_F)\, Z(s, W, \Phi).$$

For generic π, Shahidi's theorem gives $\gamma \neq 0$, so $Z(s, W, \Phi) \not\equiv 0$.

Diagnosed failure mode: The "misaligned anchor" = using ψ_F instead of $\psi_E = \psi_F \circ \mathrm{Tr}_{E/F}$ in the Whittaker model. This shifts the γ-factor normalization, making the nonvanishing argument circular. Fix: always build $\mathcal{W}(\pi, \psi_E)$ with ψ_E.

[PARTIAL — gap logged]
Construction: Unramified $\pi \to W^\circ$; ramified $\pi \to W_\mathfrak{f}$ (newform in $\mathcal{W}(\pi, \psi_E)$). Nonvanishing from Asai L-function + functional equation.
Gap: Full proof requires verifying local Langlands compatibility of the twisted character with the Asai lift (Anandavardhanan–Rajan / Matringe local theory).

3 P3: A Markov chain from interpolation polynomials

Statement. Random points on $[0, 1]$; interpolation polynomials through them define a Markov chain on coefficients. Prove convergence and identify the stationary distribution as $n \to \infty$.

SMIC log. Genesis: domain `PROB`, phase **Propagate**.
`C5:Misfit (mild)` — the precise Markov chain transition kernel is not specified in the problem. A canonical interpretation is adopted below.

3.1 Step 1 — *repr_switch*: Divided-difference state space

State at step n: the divided-difference table $(f[x_{i_0}, \ldots, x_{i_k}])_{k \geq 0}$. Adding a new uniform random point $x_{n+1} \in [0, 1]$ updates via Newton's identity:

$$f[x_0, \ldots, x_{n+1}] = \frac{f[x_1, \ldots, x_{n+1}] - f[x_0, \ldots, x_n]}{x_{n+1} - x_0}.$$

This is a local update; the chain has bounded transition cost.

3.2 Step 2 — *comb_invariant_potential*: Monovariant

Monovariant: $\mathbb{E}\|P_n - f\|_{L^2}^2$ is non-increasing. Convergence $\|P_n - f\|_{L^2} \to 0$ in probability follows from the Erdős–Turán theorem for uniform random nodes.

3.3 Step 3 — *proof_induction*: Stationary distribution

By a CLT for divided differences (when f smooth), the joint law of $(f[x_0, \ldots, x_k])_{k=0}^m$ converges to a Gaussian process with covariance determined by the Green's function $G_{[0,1]}$ of the unit interval.

[PARTIAL — gap logged]
Proposed answer: Stationary distribution = Gaussian process on coefficient space with covariance $G_{[0,1]}$.
Gap: Problem statement does not fully specify the Markov chain. Rate: geometric for analytic f, polynomial for $f \in C^k$.

4 P4: Finite additive convolution and a harmonic-mean inequality for Φ_n

Statement. Let Φ_n be the n-th cyclotomic polynomial. Establish a harmonic-mean inequality for the finite additive convolution operator acting on the space of roots.

SMIC log. Genesis: domain `NT`, phase **Integrate**.
`C5:Misfit FIRED at Scale 2.` "Additive convolution on the space of roots of Φ_n" is not a standard mathematical object. The channel fires because the operator is undefined.

4.1 Step 1 — *algebra_substitute*: Roots and Ramanujan sums

Roots of Φ_n: $\{\zeta_n^k : 1 \leq k \leq n,\ \gcd(k, n) = 1\}$, count $= \varphi(n)$. Harmonic mean:

$$\mathrm{HM}(\Phi_n) = \frac{\varphi(n)}{\displaystyle\sum_{\gcd(k,n)=1} \zeta_n^{-k}} = \frac{\varphi(n)}{\mu(n)},$$

using $\sum_{\gcd(k,n)=1} \zeta_n^k = \mu(n)$ (Ramanujan sum). For $n > 2$: $|\mu(n)| \leq 1$, so the harmonic mean is not bounded uniformly.

4.2 Step 2 — *repr_switch*: Three interpretations of "additive convolution"

A — Coefficient convolution of Φ_n, Φ_m as polynomials (Cauchy product) $\to$ coefficients of $\Phi_n \cdot \Phi_m$.
B — Sumset of root sets: $\{\alpha + \beta : \alpha \in \mathrm{roots}(\Phi_n),\ \beta \in \mathrm{roots}(\Phi_m)\}$.
C — A Mahler-measure analog: $M(\Phi_n \cdot \Phi_m) = M(\Phi_n) \cdot M(\Phi_m)$ (multiplicative, known). An HM analog is not standard.

Under none of these interpretations is the inequality straightforward to state or prove.

[OPEN — C5:Misfit]
What holds: For n prime, all roots on unit circle; AM = HM = 1; AM≥HM is trivially tight.
Gap: Cannot proceed without a precise definition of "additive convolution on root space". `C5:Misfit` logged. Recommendation: request clarification from problem setter.

5 P5: The O-adapted slice filtration and a geometric fixed-point criterion

Statement. Construct the O-adapted slice filtration in equivariant stable homotopy theory. Prove a geometric fixed-point criterion for slice connectivity of a G-spectrum.

SMIC log. Genesis: domain `TOP`, phase **Transform** (high-complexity TRANS). Scale ladder: Scale 1 (HHR framework) → Scale 3 (induction on subgroups).

5.1 Step 1 — *domain_switch*: HHR slice filtration

Hill–Hopkins–Ravenel (2016) provide a tower $\cdots \to P^{n+1}X \to P^nX \to \cdots \to X$ with slice sections $s_nX := P^nX/P^{n+1}X$.

O-adapted extension: Replace the $\mathbb{Z}$-grading with $RO(G)$ (real representation ring). Define P^VX for $V \in RO(G)$; slice sections s_VX are S^V-indexed $H\underline{\mathbb{Z}}$-module spectra.

5.2 Step 2 — *repr_switch*: Geometric fixed points

The geometric fixed-point functor:

$$\Phi^H X \;\simeq\; \left(\widetilde{E\mathcal{P}_H} \wedge X\right)^H,$$

where $\mathcal{P}_H$ is the family of proper subgroups of H.

5.3 Step 3 — *proof_extremal*: Criterion

Proposed criterion: X is slice-n-connected if and only if

$$\Phi^H(X) \text{ is } \left\lfloor \frac{n}{|W_GH|} \right\rfloor\text{-connected} \quad \forall\, H \le G,$$

where $W_GH = N_G(H)/H$.

Proof strategy: (i) Tom Dieck splitting decomposes $\Sigma^\infty_+(G/H_+)$. (ii) The $E\mathcal{F}$-long exact sequence propagates connectivity between fixed-point levels. (iii) Induction on subgroups (well-founded, G compact finite) closes the argument.

[PARTIAL — gap logged]
Framework: HHR slice filtration + $RO(G)$-grading established. Criterion: slice-n-connected ⇔ Φ^H is $\lfloor n/|W_GH| \rfloor$-connected $\forall H$.
Gap: Full verification requires working through the $E\mathcal{F}$-spectral sequence and the Weyl-group normalization — heavy equivariant stable homotopy machinery.

6 P6: Large ε-light vertex subsets

Statement. $G = (V, E)$ graph, average degree d. $S \subseteq V$ is ε-light if $\rho(A_S) < \varepsilon$. Prove $\exists\, S$ ε-light with $|S| \ge f(\varepsilon, d)|V|$.

SMIC log. Genesis: domain `COMB`, phase **Decay** (extremal). Scale 0: independent set observation. No failure channels triggered.

6.1 Step 1 — *proof_extremal*: Independent set is 0-light

Any independent set I satisfies $A_I = 0$, hence $\rho(A_I) = 0 < \varepsilon$ for all $\varepsilon > 0$. So it suffices to find a large independent set.

Caro–Wei bound: Every graph with average degree d contains an independent set of size

$$|I| \geq \sum_{v \in V} \frac{1}{d_v + 1} \geq \frac{|V|}{d+1},$$

where the second inequality uses convexity of $x \mapsto 1/(x+1)$.

6.2 Step 2 — *nt_mod_invariant*: Stronger bound for large ε

For $\varepsilon > C\sqrt{d}$: a uniformly random subset S of size $|V|/2$ has $\mathbb{E}[\rho(A_S)] \leq O(\sqrt{d})$ by the Wigner semicircle law. By a concentration argument, $\exists\, S$ with $|S| = |V|/2$ and $\rho(A_S) < \varepsilon$.

6.3 Step 3 — Complete proof and tight bound

Proof. Let $I \subseteq V$ be a maximal independent set (exists by Zorn's lemma, or by greedy construction in polynomial time). By Caro–Wei, $|I| \geq |V|/(d+1)$. Since I is independent, A_I is the zero matrix, so $\rho(A_I) = 0 < \varepsilon$. Set $S = I$ and $f(\varepsilon, d) = 1/(d+1)$. □

Tight bound combining both regimes:

$$f(\varepsilon, d) = \max\left(\frac{1}{d+1}, \frac{\varepsilon^2}{d+\varepsilon^2}\right).$$

Matching upper bound: The complete graph K_{d+1} shows no subset of size $\omega(|V|/d)$ can have spectral radius $o(d)$.

[SKETCH COMPLETE]
$f(\varepsilon, d) = 1/(d+1)$ proved via Caro–Wei. Tight bound: $\max\big(1/(d+1), \varepsilon^2/(d+\varepsilon^2)\big)$. Scale 3. No failure channels.

7 P7: Uniform lattices with 2-torsion in closed manifolds

Statement. Prove existence of uniform lattices containing 2-torsion that arise as fundamental groups of closed manifolds with acyclic universal covers.

SMIC log. Genesis: domain `TOP`, phase **Emerge**. `C4:AnchorDrift FIRED at Step 3.` The three conditions (uniform lattice, 2-torsion, manifold π_1, acyclic cover) are in fundamental tension.

7.1 Step 1 — *proof_extremal*: Arithmetic hyperbolic lattices

Arithmetic lattices in $\mathrm{SO}(n,1)$ are uniform (cocompact) and can contain involutions (reflections of order 2). Their orbit space $M = \mathbb{H}^n/\Gamma$ has $\pi_1(M) = \Gamma$, and $\mathbb{H}^n$ is contractible, hence acyclic.

7.2 Step 2 — *repr_switch*: The torsion–manifold obstruction

Obstruction detected: A group acts freely and properly discontinuously on a contractible space $\widetilde{M}$ iff it is torsion-free (Killing–Hopf). If Γ has 2-torsion, the action on $\mathbb{H}^n$ has fixed points, and $M = \mathbb{H}^n/\Gamma$ is an *orbifold*, not a manifold.

Selberg's lemma gives a torsion-free finite-index subgroup $\Gamma_0 < \Gamma$, yielding a genuine closed manifold $M_0 = \mathbb{H}^n/\Gamma_0$; but $\pi_1(M_0) = \Gamma_0$ is then torsion-free, losing the 2-torsion.

7.3 Step 3 — `C4:AnchorDrift` and possible resolutions

`C4` fires: the problem conditions appear to conflict with Killing–Hopf for the manifold/torsion combination.

Possible resolutions:

1. The problem means the orbifold category (standard in geometric group theory).
2. "Uniform lattice" refers to a lattice in a locally compact group, not necessarily acting on the universal cover of M.
3. The problem uses "acyclic" in a weaker sense (e.g., rational coefficients).

What is established: Existence of uniform lattices with 2-torsion in $\mathrm{SO}(n,1)$ acting on contractible $\mathbb{H}^n$ is classical (Borel 1963, Bergeron–Haglund–Wise).

[PARTIAL — gap logged] `C4:AnchorDrift` logged.
Gap: "Closed manifold" with torsion π_1 and contractible universal cover requires orbifold category or a non-standard reading of the problem. The potential inconsistency is flagged for the problem setter.

8 P8: Smoothability of quadrivalent polyhedral Lagrangian surfaces

Statement. Prove every quadrivalent polyhedral Lagrangian surface in (M^4, ω) is Lagrangian-smoothable via continuous deformation.

SMIC log. Genesis: domain `GEO` (symplectic), phase **Transform** (PL $\to$ smooth category switch). Scale 1 $\to$ Scale 3.

8.1 Step 1 — *domain_switch*: Local model at a quadrivalent vertex

At each quadrivalent vertex v: four edges meet. Local symplectic model: $(\mathbb{R}^4, \omega_{\mathrm{std}})$ with $\omega_{\mathrm{std}} = dx_1 \wedge dy_1 + dx_2 \wedge dy_2$. The four faces pair into two Lagrangian 2-planes meeting at v: $L_1 = \{y_1 = y_2 = 0\}$, $L_2 = \{x_1 = x_2 = 0\}$.

8.2 Step 2 — *repr_switch*: Maslov index as smoothing obstruction

The smoothing obstruction at v is the Maslov index $\mu(L_1, L_2) \in \mathbb{Z}$. Smoothability $\Leftrightarrow$

$$\sum_{\text{vertices } v} \mu(L_{1,v}, L_{2,v}) = 0.$$

8.3 Step 3 — *proof_contradiction*: Maslov balancing

Claim: The Lagrangian condition on all four faces at a quadrivalent vertex forces $\mu(L_{1,v}, L_{2,v}) = 0$.

Argument: The symplectic frame traces a loop in Sp(4) as one traverses the link of v. The Maslov index equals the winding number of this loop around the Maslov cycle in Sp(4). A contractible loop has winding number 0; the quadrivalent Lagrangian structure forces contractibility by the balancing of the four face angles.

8.4 Step 4 — *proof_extremal*: h-principle conclusion

With vanishing local obstruction, the Eliashberg–Mishachev h-principle for Lagrangian immersions applies. A Lagrangian isotopy deforms L to a smooth Lagrangian. Embeddedness follows from the Lagrangian isotopy extension theorem.

[PARTIAL — gap logged]
Sketch: Maslov balancing at quadrivalent vertices + h-principle $\Rightarrow$ smoothability.
Gap: (i) Claim "quadrivalent $\Rightarrow \mu = 0$" requires a detailed PL symplectic argument not reproduced here. (ii) h-principle gives immersion; embeddedness needs separate argument.

9 P9: Algebraic relations among scaled quadri-linear determinant tensors

Statement. Determine the complete set of algebraic relations among normalized scaled quadri-linear determinant tensors. Show these generate the invariant theory for the tensor structure.

SMIC log. Genesis: domain `ALG` (invariant theory), phase **Integrate**. Scale 1 → Scale 3 (Weyl FFT+SFT). `C5:Misfit (mild)` — "scaled/normalized" not standardly defined.

9.1 Step 1 — *algebra_substitute*: Quadri-linear determinant tensor

Let $V = k^4$ (k algebraically closed), $v_1, \ldots, v_m \in V$. The quadri-linear determinant tensor:

$$D_{i_1 i_2 i_3 i_4}(\mathbf{v}) = \det(v_{i_1}, v_{i_2}, v_{i_3}, v_{i_4}),$$

a $\mathrm{GL}_4(k)$-invariant element of $\bigwedge^4 V \cong k$. The *scaled* version $\hat{D}$ normalizes by D_{1234} (or by Hilbert–Schmidt norm).

9.2 Step 2 — *repr_switch*: First and Second Fundamental Theorems

FFT (Weyl): The invariant ring $k[(k^4)^m]^{\mathrm{GL}_4}$ is generated by the 4×4 minors $\{D_{i_1 i_2 i_3 i_4}\}$.

SFT (Weyl): The ideal of relations is generated by the Plücker relations:

$$\sum_{\sigma \in S_3} (-1)^{\mathrm{sgn}(\sigma)} D_{i_{\sigma(1)} i_{\sigma(2)} j_1 j_2} D_{i_{\sigma(3)} j_3 j_4 j_5} = 0,$$

defining the Plücker ideal of $\mathrm{Gr}(4, m)$.

9.3 Step 3 — *proof_extremal*: Scaled version and generation

Scaling $D \to \hat{D} = D/D_{1234}$ (rational normalization):

1. Plücker relations among $D_{i_1 i_2 i_3 i_4}$ hold unchanged.
2. Normalization: $\hat{D}_{1234} = 1$.
3. These generate $\mathcal{O}(U_{1234})$ where $U_{1234} = \{D_{1234} \neq 0\} \subset \mathrm{Gr}(4, m)$.

The coordinate ring $\mathcal{O}(\mathrm{Gr}(4, m)) \cong k[\hat{D}_{i_1 i_2 i_3 i_4}]/(\text{Plücker})$ equals the invariant ring $k[(k^4)^m]^{\mathrm{SL}_4}$ by the FFT.

[PARTIAL — gap logged]
Framework: Classical Weyl FFT+SFT gives complete relations for 4×4 minors.
Gap: "Scaled quadri-linear" is not a standard term. If normalization is non-algebraic (Hilbert–Schmidt), the invariant ring changes. Problem setter should clarify the scaling convention.

10 P10: Kernelized CP-ALS subproblem with missing data

Statement. Solve the kernelized CP-ALS subproblem with missing data. Construct a matrix-free PCG algorithm with Kronecker preconditioning. Prove superlinear convergence.

SMIC log. Genesis: domain `ALG` (numerical linear algebra), phase **Propagate**. Scale 0 (normal equations) → Scale 3 (Greenbaum–Strakoš).

10.1 Step 1 — *algebra_substitute*: ALS subproblem formulation

CP decomposition: $\mathcal{T} \approx \sum_{r=1}^{R} \mathbf{a}_r \otimes \mathbf{b}_r \otimes \mathbf{c}_r$. Fixing B, C, mode-1 ALS with missing entries Ω:

$$\min_A \sum_{(j,k)\in\Omega} \Big(t_{ijk} - \textstyle\sum_r a_{ir} b_{jr} c_{kr}\Big)^2.$$

Normal equations: $M_\Omega \operatorname{vec}(A) = \mathbf{y}_\Omega$, where $M_\Omega = \sum_{(j,k)\in\Omega} (c_k \otimes b_j)(c_k \otimes b_j)^T$.

10.2 Step 2 — *repr_switch*: Kernelization and Kronecker preconditioner

Kernel substitution $\mathbf{a}_r \leftarrow K_A^{1/2} \boldsymbol{\alpha}_r$:

$$\big(K_A \odot_\Omega K_B \odot_\Omega K_C + \lambda I\big)\boldsymbol{\alpha} = \mathbf{y}.$$

Kronecker preconditioner: $P = K_A \otimes K_B \otimes K_C$. Applied matrix-free:

$$P^{-1}\mathbf{v} = (K_A^{-1} \otimes K_B^{-1} \otimes K_C^{-1})\mathbf{v},$$

computed via three successive solves using the eigendecompositions $K_X = U_X \Lambda_X U_X^T$ (precomputed once per ALS sweep, cost $O(n^3)$ per mode).

10.3 Step 3 — *proof_induction*: Superlinear convergence

Standard PCG bound: $\|e_k\| \le 2\big(\frac{\sqrt{\kappa}-1}{\sqrt{\kappa}+1}\big)^k \|e_0\|$, $\kappa = \kappa(P^{-1}A)$.

Superlinear regime (Greenbaum–Strakoš 1992): PCG converges superlinearly when eigenvalues of $P^{-1}A$ cluster. Decompose:

$$P^{-1}A = I + P^{-1}(A - P) = I + E_\Omega,$$

where E_Ω captures the mask perturbation. As $|\Omega| \to |\Omega_{\max}|$ (full observation): $\|E_\Omega\|_F \to 0$, eigenvalues cluster at 1, and PCG exhibits superlinear convergence.

Rate: $\|e_k\| = O\big(\exp(-c\sqrt{k})\big)$ for $c > 0$ depending on the spectral gap of E_Ω.

[PARTIAL — gap logged]
Algorithm: Matrix-free PCG with Kronecker preconditioner constructed. Superlinear convergence from Greenbaum–Strakoš + eigenvalue clustering.
Gap: (i) Requires non-degeneracy of CP decomposition (swamping can stall convergence). (ii) Quantitative bound on $\|E_\Omega\|_F$ depends on mask pattern and is problem-

instance specific.

Summary

#	Problem	Scale	Verdict	Channels fired
1	Φ^4_3 on $\mathbb{T}^3$	3	PARTIAL	`C6`
2	Twisted RS nonvanishing	3	PARTIAL	anchor noted
3	Markov / interp poly	2	PARTIAL	`C5 mild`
4	Cyclotomic HM inequality	2	OPEN	`C5`
5	O-adapted slice	3	PARTIAL	—
6	ε-light subsets	3	SOLVED	—
7	Uniform lattices, 2-torsion	3	PARTIAL	`C4`
8	Polyhedral Lagrangian	3	PARTIAL	—
9	Quadri-linear det tensors	3	PARTIAL	`C5 mild`
10	Kernelized CP-ALS + PCG	3	PARTIAL	—

Honest assessment. Problem 6 is the only problem admitting a complete proof within reach: it reduces to the classical Caro–Wei bound, a tool fully within the combinatorics module.

For the remaining nine, SMIC provides a framework, a technique route, a partial argument, and an explicit gap log. The system does not fabricate proofs. It navigates to the boundary of what is accessible and stops there.

Problems 4 and 7 illustrate the failure-detection value of the SMIC channels: Problem 4 reveals an ambiguous statement (`C5:Misfit`); Problem 7 reveals a potential internal inconsistency (`C4:AnchorDrift`). Both would be invisible to a system that simply tries to produce an answer.